$f(x) = x|x|$ is also an increasing function.

Horizontal Test

Similar to the vertical line test we have a test called horizontal test to check whether a function is one-to-one, onto or not. Let a function be given as a curve in the plane. If the horizontal line through a point y in the co-domain meets the curve at some points, then the x-coordinate of all the points give pre-images for y.

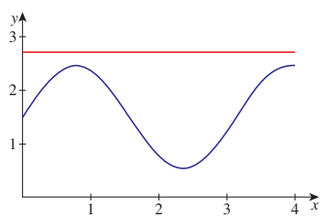


Figure 1.32

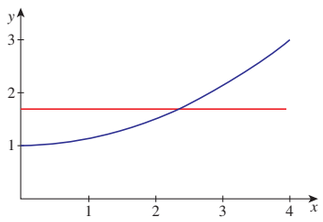


Figure 1.33

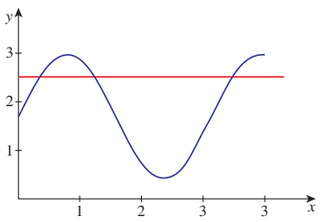


Figure 1.34

(i) If the horizontal line through a point y in the co-domain does not meet the curve, then there will be no pre-image for y and hence the function is not onto.
(ii) If the horizontal line through atleast one point of the codomain meets the curve at more than one point, then the function is not one-to-one.
(iii) If for all y in the range the horizontal line through y meets the curve at only one point, then the function is one-to-one.

So we may say, the function represented by a curve is one-to-one if and only if for all y in the range, the horizontal line through the point y meets the curve at exactly one point.

The function represented by a curve is onto if and only if for all y in the co-domain, the horizontal line through the point y meets the curve atleast one point.

The curve given in Figure 1.32 represents a function from $[0, 4]$ which is not onto if the co-domain contains $[1, 3]$. The curve given in Figure 1.33 represents a one-to-one function from $[0, 4]$ to $\mathbb{R}$ and the curve given in Figure 1.34 represents a function from $[0, 4]$ to $\mathbb{R}$ which is not one-to-one.

Testing whether a given curve represents a one-to-one function, onto function or not by drawing horizontal lines is called ***horizontal line test*** or simply horizontal test.

Further by seeing the diagrams in Illustration 1.2 and Figures 1.5 to Figure 1.7, the function

(i) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x$ is an one-to-one and onto function.
(ii) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is neither one-to-one nor onto.
(iii) $f : [0, \infty) \to \mathbb{R}$ defined by $f(x) = +\sqrt{x}$ is an one-to-one but not onto function.
(iv) $f : [0, \infty) \to [0, \infty)$ defined by $f(x) = +\sqrt{x}$ is an one-to-one and onto function.
(v) $f : [0, \infty) \to \mathbb{R}$ defined by $f(x) = -\sqrt{x}$ is one-to-one but not onto function.
(vi) $f : [0, \infty) \to (-\infty, 0]$ defined by $f(x) = -\sqrt{x}$ is one-to-one and onto function.

Example 1.21 Find the largest possible domain for the real valued function f defined by $f(x) = \sqrt{x^2 - 5x + 6}$.

Solution:

As we are finding the square root of $x^2 - 5x + 6$, we must have $x^2 - 5x + 6 \geq 0$ for all x in the domain. For this, follow the given procedure.

Solving $x^2 - 5x + 6 = 0$, we get $x = 2$ and 3. Now draw the number line as in Figure 1.35.

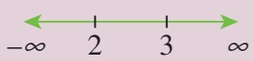


Figure 1.35